Bluetooth Based Attendance System - Project Synopsis

Abstract

This project automates attendance by scanning nearby Bluetooth devices and matching detected device identifiers against a registered student database. When a match is found, attendance is recorded with the current date and time.

Objectives

• Reduce manual attendance.
• Maintain digital attendance records.
• Prevent duplicate attendance for the same day.
• Demonstrate Python automation with Streamlit.

Technology Stack

Python, Streamlit, Pandas, Bleak (Bluetooth LE), CSV files.

Modules

1. app.py - User interface.
2. bluetooth_scanner.py - Scans nearby Bluetooth devices.
3. database.py - Loads registered students.
4. attendance.py - Marks attendance and avoids duplicates.

Working

User clicks 'Scan Bluetooth Devices'. Nearby Bluetooth devices are detected. Their identifiers are compared with the registered student database. If a match is found, attendance is saved in attendance.csv with date and time.

Advantages

Fast, paperless, easy to use, automatic record keeping.

Limitations

Modern mobile operating systems may hide or randomize Bluetooth MAC addresses, so this implementation is best suited as a proof-of-concept.

Future Enhancements

Faculty login, student registration, attendance dashboard, downloadable reports, cloud database integration, QR/RFID support.